Navigation models: cost, decision time and driving

Four robots, shared policy, safety layer on. Driving figures: 400 robots on held-out test layouts 100–199 (2D). No success or collision figures on this page; those are in benchmark/RESULTS.md.

qaro swarm_nav_rl
benchmark 2026-11-10 · arenas 9001–9005
2026-11-11

RESOURCES AND DRIVING

Model	Training / tuning cost	Model size	Decision time 2D / Gazebo, median	Avg speed	Path efficiency[1]	Time to goal[2]	Steering flips / min	Time stopped
PPO	8M steps, about 2 h, laptop CPU original project	92,717 weights 372 KB	113 µs / 98 µs	0.127 m/s	0.97	10.9 s	**218**	4%
DQN seed 2	6M steps, 1.7 h, Colab T4 GPU about 960 steps/s; tuning 6 × 1M steps, 1.0 h	92,717 weights 372 KB	83 µs / 84 µs	0.090 m/s	0.96	16.1 s	**249**	10%
Rule controller	No training; 50 parameter sets tuned 1.7 h (4.3 CPU-h)	7 parameters	160 µs / 1.25 ms	0.085 m/s	0.99	9.6 s	12	1%
Tabular Q seed 3	13 training runs 6.4 h (6.3 CPU-h)	291,600-entry table 1.2 MB	144 µs / –	0.087 m/s	0.86	15.9 s	204	12%
DWA privileged[3]	None	none	not timed needs simulator state	0.066 m/s	0.93	11.9 s	4	0%
Go-to-goal	None	none	22 µs / –	0.118 m/s	1.07[4]	8.9 s	14	7%

[1] Straight-line distance / distance driven (1.0 = perfectly direct). [2] Mean over robots that reached their goal. [3] Uses the true map of the arena, so it cannot be deployed on the real robots. [4] Can exceed 1: a robot may stop up to 0.15 m short of the goal centre. Every model decides in under 2% of the 100 ms decision period, so computation is not a limit for any of them.

VERDICT

Model	Normal arenas	Adversarial arenas (9001–9005: robots must cross)
PPO	Among the best: fast and direct, but its steering wobbles constantly.	Fine at open crossings; breaks down at doors and narrow gaps, and crashes into robots without the safety layer.
DQN	On par with PPO but more cautious: slower, stops more, crashes less, wobbles as much.	Held up best in 2D; when it fails it gets stuck rather than crashes.
Rule controller	Slightly behind the learned policies; the smoothest, steadiest driver, but gets trapped behind clusters of obstacles.	Good in corridors and at the door in 2D, but deadlocks around a central pillar and stalls in clutter.
Tabular Q	Clearly weaker: indirect paths, frequent obstacle hits.	Fails almost everywhere except the easy crossroads.
DWA privileged	Decent, very smooth but slow; not deployable (needs the true map).	Stalls often, even with the true map.
Go-to-goal	Baseline only: crashes whenever something is in the way.	Fails almost everywhere.

Caveats: PPO's training cost is from the original project's documentation, not measured in the benchmark. DQN's cost is for the one seed kept; three seeds were trained at the same cost each. Adversarial verdicts rest on one episode per arena, mostly in 2D (Gazebo checks only for PPO): impressions, not measurements.